



Step 1. Task Setup

Task type	pick_clean_then_place_in_recep
User goal	Put a clean apple on the diningtable.
Initial observation	You are in the kitchen. You see a countertop, a sinkbasin, a fridge, acabinet, and a diningtable. You are notholding anything.
Target terminal condition	The apple is clean and placed on the diningtable.

Initial admissible actions:

- look
- inventory
- go to countertop 1
- go to sinkbasin 1
- go to fridge 1
- go to cabinet 1
- go to diningtable 1

Step 2. Architect Blueprint Generation

Input
task_goal
initial_observation
available_actions

Output
Schema includes: task_goal, current_observation, inventory,available_actions, plan, candidate_action, verification,repair_hint, action_history, workflow_status
Worker contracts: planner, actor, verifier, repair
Workflow rules: initial_state → planner → actor → verifier→ environment_step / repair



Committed effect

workflow_status = planning
plan.target_object = apple
plan.target_receptacle = diningtable
plan.status = draft



Step 3. Planner Patch

```
{ "op": "replace", "path": "/plan/subgoal",  
  "value": "Find an apple, clean it at the sinkbasin, then place it on the diningtable." },  
{ "op": "replace", "path": "/plan/target_object", "value": "apple" },  
{ "op": "replace", "path": "/plan/target_receptacle", "value": "diningtable" },  
{ "op": "replace", "path": "/plan/status", "value": "active" },  
{ "op": "replace", "path": "/workflow_status", "value": "acting" }
```



Kernel result:

Accepted. Paths are authorized;
state validates.



Step 4. Main Execution Loop

Turn 1



Actor

Action: go to countertop 1

Reason: Countertops commonly contain apples.



Verifier:

admissible = true
progressing = true



Environment:

You are at countertop 1.
You see apple 1 on countertop 1.

Turn 2



Actor

Action: take apple 1 from countertop 1



Verifier:

accepted



Environment:

You pick up apple 1.
Inventory: apple 1

Turn 3



Actor

Action: go to sinkbasin 1



Verifier:

accepted



Environment:

You are at sinkbasin 1.

Turn 4



Actor

Action: clean apple 1 with sinkbasin 1



Verifier:

accepted



Environment:

The apple 1 is now clean.
Inventory: clean apple 1



Step 5. Repair / Retry Event



Initial actor proposal:

put apple 1 on diningtable 1



Verifier output:

is_admissible = false
is_progressing = false
needs_repair = true

comment: The agent is not at the diningtable, so theproposed put action is not currently admissible.



Repair worker patch:

repair_hint = Navigate to diningtable 1 before placing the clean apple.
plan.status = revised
workflow_status = acting



Kernel note:

Rejected action is logged. Repair hint is accepted and committed.
Shared stateis not contaminated by the inadmissible action.



Retry:

Retried actor patch: go to diningtable 1
Verifier accepts. Environment: You are at diningtable 1.



Step 6. Final Action and Committed State

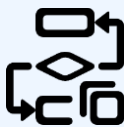
Final actor patch: put apple 1 on diningtable 1

Final verifier:

is_admissible = true
is_progressing = true
needs_repair = false

Environment:

Task complete.
Terminal success: true



```
workflow_status = completed  
inventory = []  
plan.target_object = apple  
plan.target_receptacle = diningtable  
plan.status = done  
success = true  
action_history = [  
  "go to countertop 1",  
  "take apple 1 from countertop 1",  
  "go to sinkbasin 1",  
  "clean apple 1 with sinkbasin 1",  
  "go to diningtable 1",  
  "put apple 1 on diningtable 1"  
]
```